PHYS 365 — Final Exam Fall 2022 — Worked Solutions

30 questions. Every step uses the formula sheet printed at the back of the F2022 final. Anything that is *not* on that sheet is put in an orange **REMEMBER** box. Every final answer was checked against the official answer key ("phys 365 final_exam_answer key.pdf"). That key is a slightly different printing of this exam. The letters below are for the **printed exam** shown in each question image.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	6	C	11	D	16	A	21	C	26	none*
2	C	7	C	12	B	17	D	22	B	27	B
3	C	8	D	13	C	18	C	23	D†	28	B
4	A	9	C	14	A	19	C	24	A	29	D
5	D	10	C	15	B	20	A	25	B	30	C

NOTE ON THE ANSWER KEY

*Q26: the options are printed with the wrong units. The key itself says the correct answer is 2.73×10^{-22} kg·m·s^{-1}. †Q23: the key's version lists the same answer as option A. In this printed exam it is option D.

REMEMBER — not on the formula sheet

- Same μ: $\tau = \mu v^2$, so $\tau \propto v^2$ (Q1).
- Higher tension means a faster wave, so fewer loops are needed for the same frequency (Q3).
- Four equal phasors spaced $\frac{\pi}{2}$ apart cancel (Q4).
- $a = -\omega^2 x$ (Q5).
- Small angles: $y_m = \dfrac{m\lambda D}{a}$ (Q9).
- Largest possible path difference between two sources is d (Q11).
- Number of maxima around two sources: $|m| < \frac{d}{\lambda}$ gives 2 points each, and $|m| = \frac{d}{\lambda}$ gives 1 point each (Q12).
- Thin film, transmitted rays, no net reflection shift: maxima $2L = m\dfrac{\lambda}{n_2}$, minima $2L = \left(m + \frac{1}{2}\right)\dfrac{\lambda}{n_2}$ (Q13, Q14).
- Electron rest energy: $E_0 = m_e c^2$ (Q26).
- Photon: $E = \dfrac{hc}{\lambda}$. Particle: $\lambda = \dfrac{h}{p} = \dfrac{h}{\sqrt{2mK}}$ (Q24, Q28, Q29).

Question 1 Ranking tensions

1. The following four waves are sent along strings with the same linear densities (x is in meters and t is in seconds). Rank the tension (τ) in the strings along which they travel, greatest first:

$$y_1 = (3\text{ mm})\sin(x - 3t)$$
$$y_2 = (6\text{ mm})\sin(2x - t)$$
$$y_3 = (1\text{ mm})\sin(4x - t)$$
$$y_4 = (2\text{ mm})\sin(x - 2t)$$

A) $\tau_1 > \tau_4 = \tau_2 > \tau_3$
B) $\tau_3 > \tau_2 = \tau_4 > \tau_1$
C) $\tau_3 > \tau_2 > \tau_4 > \tau_1$
D) $\tau_1 > \tau_4 > \tau_2 > \tau_3$

FROM THE FORMULA SHEET

$$v = \frac{\omega}{k}, \quad v = \sqrt{\frac{F_T}{\mu}}$$

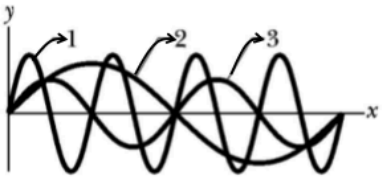
$$v_1 = \frac{3}{1} = 3, \qquad v_2 = \frac{1}{2}, \qquad v_3 = \frac{1}{4}, \qquad v_4 = \frac{2}{1} = 2$$

Since $\tau = \mu v^2$ with the same μ: $\tau_1 > \tau_4 > \tau_2 > \tau_3$

Final answer: **(D)** $\tau_1 > \tau_4 > \tau_2 > \tau_3$ ✓ Matches answer key

Question 2 Matching phases to snapshots

2. The figure is a composite of three snapshots, each of a wave traveling along a particular string. The phases for the waves are given by **(a) 2x − 4t, (b) 4x − 8t**, and **(c) 8x − 16t.** Which phase corresponds to which wave in the figure?



A) (a) to 1; (b) to 2; (c) to 3
B) (a) to 3; (b) to 2; (c) to 1
C) (a) to 2; (b) to 3; (c) to 1
D) (a) to 3; (b) to 1; (c) to 2

FROM THE FORMULA SHEET

$$k = \frac{2\pi}{\lambda}$$

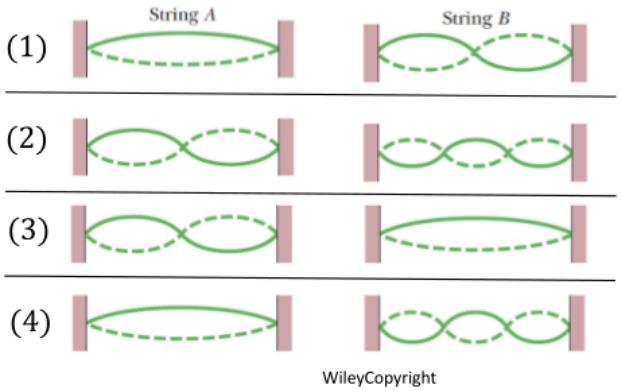
Step 1. $k = 2, 4, 8$, so $\lambda_a > \lambda_b > \lambda_c$. In the figure the longest wavelength is wave 2, then wave 3, and the shortest is wave 1.

Final answer: **(C) (a)→2, (b)→3, (c)→1** ✓ Matches answer key

Question 3 Same frequency, different tension

3. Strings A and B have identical lengths and linear densities, but string B is under greater tension than string A. Figure shows four situations, (1) through (4), in which standing wave patterns exist on the two strings. In which situations is there the possibility that strings A and B are oscillating at the same resonant frequency?

A) (1)
B) (2)
C) (3)
D) (4)



FROM THE FORMULA SHEET

$$f_n = \frac{v}{\lambda_n} = nf_1, \quad v = \sqrt{\frac{F_T}{\mu}}$$

Step 1. $v_B > v_A$. For equal frequencies $\frac{n_A v_A}{2L} = \frac{n_B v_B}{2L}$, so string B needs **fewer** loops: $n_B < n_A$.

Step 2. Only situation (3) has that (A has 2 loops, B has 1).

Final answer: **(C) (3)** ✓ Matches answer key

Question 4 Four waves 0.5π apart

4. Four waves are to be sent along the same string, in the same direction:

$$y_1(x,t) = (4.00\text{ mm})\sin(2\pi x - 400\pi t)$$
$$y_2(x,t) = (4.00\text{ mm})\sin(2\pi x - 400\pi t + 0.5\pi)$$
$$y_3(x,t) = (4.00\text{ mm})\sin(2\pi x - 400\pi t + \pi)$$
$$y_4(x,t) = (4.00\text{ mm})\sin(2\pi x - 400\pi t + 1.5\pi)$$

What is the wave function of the resultant wave of these waves?

A) $y'(x,t) = 0$
B) $y'(x,t) = (5.7\text{ mm})\sin(2\pi x - 400\pi t + 0.5\pi)$
C) $y'(x,t) = (5.7\text{ mm})\sin(2\pi x - 400\pi t)$
D) $y'(x,t) = (5.7\text{ mm})\sin(2\pi x - 400\pi t + \pi)$

FROM THE FORMULA SHEET
$\tilde{A} = Ae^{j\theta}$

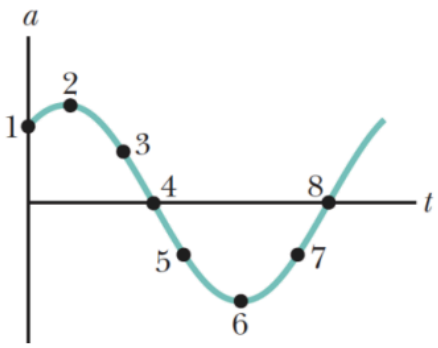
Step 1. Phasors at $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$: $4(1) + 4(j) + 4(-1) + 4(-j) = 0$. The waves cancel completely.

Final answer: **(A)** $y' = 0$ ✓ **Matches answer key**

Question 5 Position from the acceleration graph

5. The acceleration $a(t)$ of a particle undergoing SHM with amplitude x_m is graphed in the figure. At point 5, particle is at:

A) $+x_m$
B) $-x_m$
C) Between $-x_m$ and 0
D) Between 0 and $+x_m$



FROM THE FORMULA SHEET
$a = -\omega^2 x_m \cos(\omega t + \phi) = -\omega^2 x$

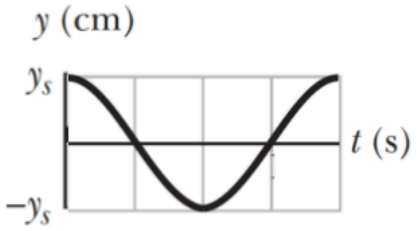
Step 1. At point 5, $a < 0$ but not at its most negative, so $x = -\dfrac{a}{\omega^2}$ is positive and less than x_m.

Final answer: **(D) Between 0 and $+x_m$** ✓ Matches answer key

Question 6 Wave function from a $y(t)$ graph

6. A sinusoidal transverse wave travels along a string in the positive direction of the x axis. The wave equation form is $y(x,t) = y_s \sin(kx - \omega t + \varphi)$. The displacement y of the string particle at $x = 0$ is given in figure as a function of time t. The wave function is to be in the form of:

A) $y(x,t) = +y_s \sin(kx - \omega t)$
B) $y(x,t) = -y_s \sin(kx - \omega t)$
C) $y(x,t) = +y_s \cos(kx - \omega t)$
D) $y(x,t) = -y_s \cos(kx - \omega t)$



FROM THE FORMULA SHEET
$y(x,t) = A\sin(kx \pm \omega t + \phi_0)$, $\quad \sin\theta = \cos\left(\theta - \frac{\pi}{2}\right)$

Step 1. At $x = 0$ the graph starts at $+y_s$ and goes down, so $y(0,t) = y_s \cos(\omega t)$.

Step 2. Try $\phi = \frac{\pi}{2}$: $y_s \sin\left(-\omega t + \frac{\pi}{2}\right) = y_s \cos(\omega t)$ ✓. Then

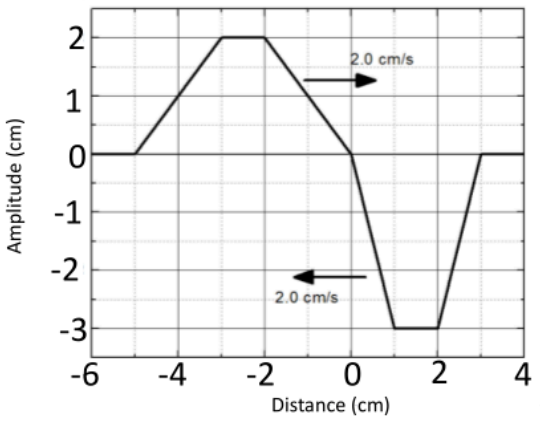
$$y = y_s \sin\left(kx - \omega t + \frac{\pi}{2}\right) = y_s \cos(kx - \omega t)$$

Final answer: **(C)** $+y_s \cos(kx - \omega t)$ ✓ Matches answer key

Question 7 Two pulses after 1 s

7. The graph shows two waves at time $t = 0$ s, one moving toward the right at 2.0 cm/s and the other moving toward the left at 2.0 cm/s. What will the amplitude be at $x = 0$ at time $t = 1.0$ s?

A) −3 cm
B) zero cm
C) −1 cm
D) −2 cm



FROM THE FORMULA SHEET
$y(x,t) = f(x \pm vt)$

Step 1. Each pulse moves $2.0 \times 1.0 = 2.0$ cm.

Step 2. The right-moving pulse brings the value it had at $x = -2$ cm, which is $+2$ cm (its flat top). The left-moving pulse brings the value it had at $x = +2$ cm, which is -3 cm (its flat bottom).

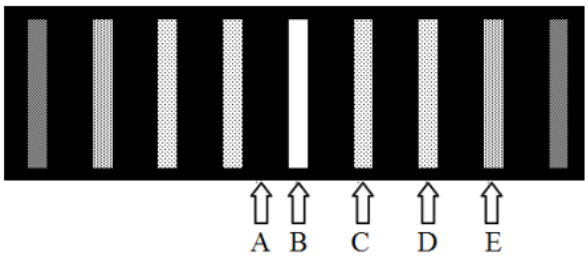
$$y = 2 + (-3) = -1 \text{ cm}$$

Final answer: **(C)** -1 **cm** ✓ **Matches answer key**

Question 8 Which fringe has a 1980 nm difference?

8. A beam of monochromatic light with a wavelength of 660 nm is directed at a double slit. Consider the five fringes labeled in the drawing, including the central maximum that is labeled "**B.**" Which one of these fringes is produced when the path length difference is 1980 nm?

A) A
B) C
C) D
D) E



FROM THE FORMULA SHEET

$\Delta = |r_2 - r_1| = m\lambda$

$$m = \frac{1980}{660} = 3$$

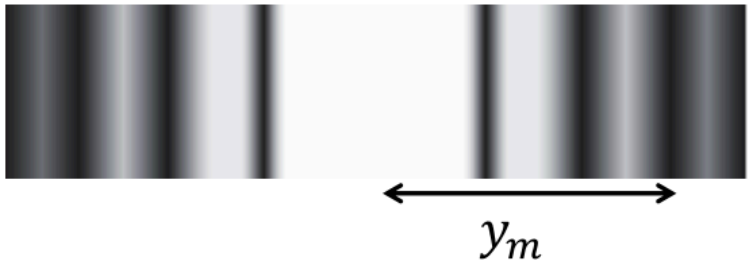
B is $m = 0$, C is 1, D is 2, so E is $m = 3$.

Final answer: **(D) Fringe E** ✓ **Matches answer key**

Question 9 Single-slit distance y_m

9. Light of wavelength 600 nm is incident upon a single slit with width 4×10^{-4} m. The figure shows the pattern observed on a screen positioned 2m from the slit. Determine the distance y_m.

A) 0.003 m
B) 0.006 m.
C) 0.009 m
D) 0.012 m



FROM THE FORMULA SHEET

$\Delta = a \sin\theta = m\lambda$

REMEMBER — not on the formula sheet

Small angles: $y_m = \dfrac{m\lambda D}{a}$

Step 1. Minima spacing: $\dfrac{\lambda D}{a} = \dfrac{(600 \times 10^{-9})(2)}{4 \times 10^{-4}} = 0.003$ m

Step 2. In the figure, y_m runs from the centre to the 3rd dark fringe: $y_3 = 3(0.003) = 0.009$ m

Final answer: **(C) 0.009 m** ✓ Matches answer key

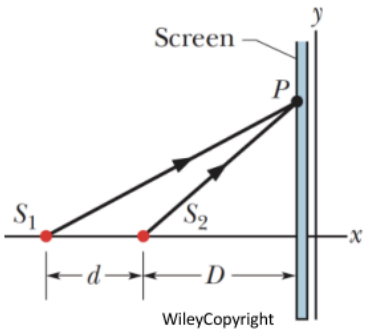
Question 10 Path difference at $y_P = d$

Questions 10 and 11 are related to the following situation:

In the Figure, two isotropic point sources S_1 and S_2 emit light in phase at wavelength λ and at the same amplitude. The sources are separated by distance $d = 6.00\,\lambda$ on an x axis. A viewing screen is at distance $D = 20.0\,\lambda$ from S_2 and parallel to the y axis. The figure shows two rays reaching point P on the screen, at height y_P.

10. What is the path length difference between two rays when $y_p = d$?

A) $3.50\,\lambda$
B) $20.0\,\lambda$
C) $5.80\,\lambda$
D) $26.7\,\lambda$



Step 1. In units of λ: S_1 is 26 back horizontally and S_2 is 20 back. The height is 6.

$$r_1 = \sqrt{26^2 + 6^2} = 26.68, \qquad r_2 = \sqrt{20^2 + 6^2} = 20.88, \qquad \Delta = 5.80\lambda$$

Final answer: **(C) 5.80 λ** ✓ Matches answer key

Question 11 Is 7.4λ possible?

11. If the path length difference between two rays from S_1 and S_2 at P (above figure) equals to $7.4\,\lambda$, the resulting intensity at point P is

A) Minimum
B) Maximum
C) Intermediate but closer to minimum
D) Path length difference $7.4\,\lambda$ for the above situation is impossible

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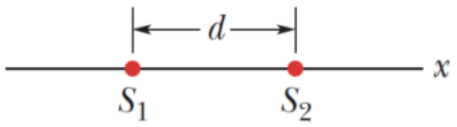
Step 1. The path difference can never be larger than $d = 6.00\lambda$, which is reached on the x-axis. So 7.4λ is impossible.

Final answer: **(D) Impossible** ✓ Matches answer key

Question 12 Counting maxima

12. In the Figure, two radiofrequency point sources S_1 and S_2, separated by distance $d = 2.0$ m, are radiating in phase with $\lambda = 0.50$ m. A detector moves in a large circular path around the two sources in a plane containing them. How many maxima does it detect?

A) 9
B) 16
C) 18
D) 20



FROM THE FORMULA SHEET
$d \sin\theta = m\lambda$

Step 1. $\dfrac{d}{\lambda} = 4$, so $m = 0, \pm1, \pm2, \pm3, \pm4$.

$$\underbrace{7 \times 2}_{|m|<4} + \underbrace{2}_{m=\pm4} = 16$$

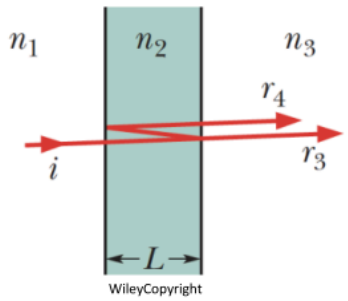
Final answer: **(B) 16** ✓ Matches answer key

Question 13 Thin film: visible minimum

Questions 13 and 14 are related to the following situation:

In the Figure, light is incident perpendicularly on a thin layer of material 2 that lies between (thicker) materials 1 and 3. (The rays are tilted only for clarity.) Part of the light ends up in material 3 as ray r_3(the light does not reflect inside material 2) and r_4(the light reflects twice inside material 2). The waves of r_3 and r_4 interfere, and here we consider the type of interference to be either maximum (max) or minimum (min). For this situation, each following questions below refers to the indexes of refraction n_1, n_2, and n_3, the type of interference, the thin-layer thickness L in nanometers, and the wavelength λ in nanometers of the light as measured in air.

Question 13 find the wavelength that is in the visible range. (The visible wavelengths range from 0.4μm to 0.7μm)



Question	n_1	n_2	n_3	Type	L	λ
13.	1.32	1.75	1.39	min	335	

A) 498 nm
B) 480 nm
C) 469 nm
D) 455 nm

FROM THE FORMULA SHEET

$\lambda = \dfrac{\lambda_{vacuum}}{n}$. Reflection off a lower index gives 0. Off a higher index gives 0.5 wavelength.

Step 1. r_4 reflects at 2|3 ($1.75 \rightarrow 1.39$, lower index, 0) and at 2|1 ($1.75 \rightarrow 1.32$, lower index, 0). There is no net shift.

Step 2. Minimum condition:

$$2L = \left(m + \tfrac{1}{2}\right)\frac{\lambda}{n_2} \;\Rightarrow\; \lambda = \frac{2(335)(1.75)}{m + \frac{1}{2}} = \frac{1172.5}{m + \frac{1}{2}}$$

Step 3. $m = 1$ gives 782 nm (just outside the visible range). $m = 2$ gives 469 nm (visible ✓).

Final answer: **(C) 469 nm** ✓ Matches answer key

Question 14 Thin film: third least thickness

Question 14 find the third least thickness.

Question	n_1	n_2	n_3	Type	L	λ
14.	1.55	1.60	1.33	Min	3rd	512

A) 400 nm
B) 423 nm
C) 478 nm
D) 497 nm

FROM THE FORMULA SHEET

$$\lambda = \frac{\lambda_{vacuum}}{n}$$

Step 1. 2|3 ($1.60 \to 1.33$, lower index) and 2|1 ($1.60 \to 1.55$, lower index) both give 0, so there is no net shift. Minimum condition:

$$L = \frac{\left(m + \frac{1}{2}\right)\lambda}{2n_2} = \frac{\left(m + \frac{1}{2}\right)(512)}{3.20} = 160\left(m + \frac{1}{2}\right) = 80,\ 240,\ 400 \text{ nm}$$

The third value is 400 nm.

Final answer: **(A) 400 nm** ✓ Matches answer key

Question 15 Power transmission between fluids

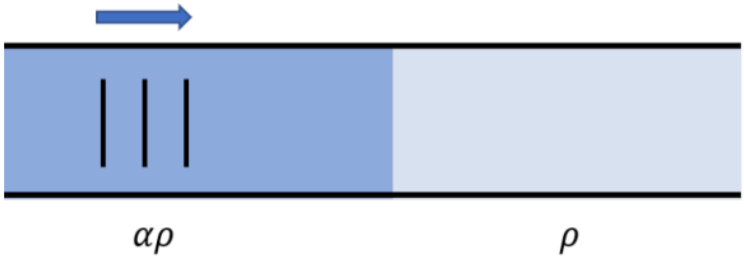
15. A wave in a fluid is incident on a boundary with another fluid. The wave travels from the left, in a fluid with mass density $\alpha\rho$, to the right, into the fluid with mass density ρ (i.e., the mass density of the fluid on the left is α times that of the fluid on the right). The two fluids have the same bulk modulus. Which option gives *power* transmission coefficients (T) at this boundary?

A) $\frac{2\sqrt{\alpha}}{1+\sqrt{\alpha}}$

B) $\frac{4\sqrt{\alpha}}{(1+\sqrt{\alpha})^2}$

C) $\frac{2}{1+\sqrt{\alpha}}$

D) $\frac{4\sqrt{\alpha}}{1+\sqrt{\alpha}}$



FROM THE FORMULA SHEET

$$Z = \sqrt{\rho B}, \quad t = \frac{2Z_1}{Z_1 + Z_2}, \quad T = \frac{Z_2}{Z_1}t^2$$

Step 1. Same B: $Z_1 = \sqrt{\alpha}\, Z_2$. Let $Z_2 = 1$:

$$T = \frac{1}{\sqrt{\alpha}}\left(\frac{2\sqrt{\alpha}}{\sqrt{\alpha}+1}\right)^2 = \frac{4\sqrt{\alpha}}{(1+\sqrt{\alpha})^2}$$

Final answer: **(B)** $\frac{4\sqrt{\alpha}}{(1+\sqrt{\alpha})^2}$ ✓ Matches answer key

Question 16 Lowest frequency from a whirling whistle

16. A whistle of frequency 540 Hz moves in a circle of radius 60 cm at an angular speed of 15.0 rad/s. What is the *lowest* frequency heard by a listener a long distance away, at rest with respect to the center of the circle? (For a circular motion the linear velocity $v = r\omega$, whereas r is the radius of the circle and ω is the angular speed.)

A) 526 Hz
B) 534 Hz
C) 555 Hz
D) 565 Hz

FROM THE FORMULA SHEET

$$f_L = \frac{v \pm v_L}{v \mp v_S} f_S$$

Step 1. Source speed (formula given in the question): $v_S = r\omega = 0.60(15.0) = 9.0\ \frac{\text{m}}{\text{s}}$

Step 2. Lowest frequency when the whistle moves away:

$$f_L = 540 \cdot \frac{343}{343 + 9.0} = 540 \cdot \frac{343}{352} = 526 \text{ Hz}$$

Final answer: **(A) 526 Hz** ✓ Matches answer key

Question 17 Photoelectric effect: FALSE statement

17. Which of the following statements is FALSE about the photoelectric effect?

A) The photoelectric effect is a consequence of the particle character of radiation.
B) There exists a minimum frequency of incident radiation below which photocurrent does not occur.
C) The work function defines the energy needed to detach a photoelectron from the surface of the target material.
D) There exists a lag time where electrons gain sufficient energy to leave the electrode surface.

FROM THE FORMULA SHEET

$$hf = K_{max} + \Phi$$

Step 1. Each photon gives its energy to one electron instantly, so there is **no** lag time. Statement D is false. A, B, and C are true.

Final answer: **(D) "There exists a lag time" is false** ✓ Matches answer key

Question 18 What is a black body?

18. What is a black body?

A) An object painted with black, non-reflective paint
B) An object that does not radiate any electromagnetic waves
C) A model of an object that emits electromagnetic waves with a characteristic frequency distribution
D) All of the above

Step 1. A black body is an idealized model: a perfect absorber that emits radiation with a spectrum set only by its temperature.

Final answer: **(C) A model that emits a characteristic frequency distribution** ✓ Matches answer key

Question 19 When is the Compton shift negligible?

19. In the Compton effect, at what scattering angle, θ would the shift in the wavelength (i.e., Compton shift) become negligible.

A) $\theta = 90^o$
B) $\theta = 45^o$
C) $\theta = 0^o$
D) $\theta = 15^o$

FROM THE FORMULA SHEET

$$\Delta\lambda = \frac{h}{mc}(1 - \cos\varphi)$$

$$\varphi = 0^\circ: \quad 1 - \cos 0 = 0 \Rightarrow \Delta\lambda = 0$$

Final answer: **(C)** 0° ✓ Matches answer key

Question 20 What the Compton effect showed

20. When the Compton shift was measured, what was demonstrated?

A) Photons have momentum.
B) Photons have energy.
C) Photons behave like waves.
D) Photons have no mass.

Step 1. The shift is explained by treating the scattering as a collision in which the photon transfers **momentum** $p = \dfrac{h}{\lambda}$ to the electron.

Final answer: **(A) Photons have momentum** ✓ Matches answer key

Question 21 Energy of an oscillating atom

21. If the frequency of vibration of an atom is f, the atom's energy is

A) Zero
B) $0.5\,hf$
C) $1.0\,hf$
D) $1.5\,hf$

FROM THE FORMULA SHEET
$E = hf$

Step 1. In Planck's model, an atom vibrating at frequency f has energy in units of hf. The key takes this as one quantum, $E = 1.0\,hf$.

Final answer: **(C)** $1.0\,hf$ ✓ Matches answer key

Question 22 Position–momentum trade-off

22. If the accuracy in measuring the position of a particle increases, the accuracy in measuring its momentum will:

A) increase.
B) decrease.
C) remain the same.
D) It is impossible to say since the two measurements are independent and do not affect each other

FROM THE FORMULA SHEET
$\Delta x\,\Delta p_x \geq \dfrac{h}{2\pi}$

Step 1. A smaller Δx forces a larger Δp, so the momentum accuracy **decreases**.

Final answer: **(B) Decreases** ✓ Matches answer key

Question 23 The classical puzzle

23. When the photoelectric effect experiments were performed, one effect was inconsistent with classical physics. What was it?

A) The fact that electrons could form a current within a vacuum.
B) The kinetic energy of the ejected electrons increased as the frequency of light increased.
C) The fact that light could free electrons from the surface of a metal.
D) The kinetic energy of the ejected electrons did not vary with light intensity.

FROM THE FORMULA SHEET

$hf = K_{max} + \Phi$

Step 1. $K_{max} = hf - \Phi$ depends on the frequency, not the intensity. Classically, brighter light should give more energetic electrons, so this result contradicted classical physics.

Final answer: **(D) KE did not vary with light intensity** **✓ Key (listed as A in the key's version)**

Question 24 Twice the photon energy

24. Photon A has twice the energy of photon B. Which statement is true?

A) the momentum of A is greater than that of B and the wavelength of A is less than that of B
B) the momentum of A is less than that of B and the wavelength of A is greater than that of B
C) the momentum of A is greater than that of B and the wavelength of A is greater than that of B
D) the momentum of A is greater than that of B and the wavelength of A equals that of B

FROM THE FORMULA SHEET

$E = pc$ (photon), $E = hf$, $p = \dfrac{h}{\lambda}$

Step 1. $E_A = 2E_B$ gives $p_A = 2p_B$ and $\lambda_A = \dfrac{h}{p_A} = \dfrac{\lambda_B}{2}$. So A has the greater momentum and the shorter wavelength.

Final answer: **(A) Greater momentum, shorter wavelength** **✓ Matches answer key**

Question 25 Stefan's law

25. Using Stefan's law, determine the total power of blackbody radiation emitted across the entire spectrum of wavelengths for a blackbody with the surface area of 25 m^2 at 60 degrees Celsius.

A) 4.72 W
B) 17.43 kW
C) 18.37 W
D) 7.45 kW

FROM THE FORMULA SHEET
$P = IA = \sigma T^4 A, \quad T_K = T_{^\circ\mathrm{C}} + 273.15$

$$P = (5.67 \times 10^{-8})(25)(333)^4 = 1.743 \times 10^4 \text{ W} = 17.43 \text{ kW}$$

(Using 333.15 K gives 17.46 kW, which is still clearly B.)

Final answer: **(B) 17.43 kW** ✓ **Matches answer key**

Question 26 Momentum of a photon with the electron's rest energy

26. What is the magnitude of the momentum associated with a photon having an energy equal to the electron rest energy?

A) $4.73\times 10^{-12}\, m$
B) $2.43\times 10^{-12}\, m$
C) $3.23\times 10^{-12}\, m$
D) $5.12\times 10^{-12}\, m$

FROM THE FORMULA SHEET
$E = pc$ (photon)

REMEMBER — not on the formula sheet
Electron rest energy: $E_0 = m_e c^2$

$$p = \frac{E}{c} = \frac{m_e c^2}{c} = m_e c = (9.109 \times 10^{-31})(3.00 \times 10^8) = 2.73 \times 10^{-22} \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}$$

NOTE ON THE ANSWER KEY
All four options are printed in metres, so none of them is correct. The key says the answer is 2.73×10^{-22} kg·m·s^{-1}.

Final answer: **(–)** 2.73×10^{-22} **kg·m·s^{-1}** ✓ **Matches key's correction**

Question 27 Doubling the position range

27. The position along the x axis of an electron is known to be between – 0.31 nm and + 0.31 nm. How would the uncertainty in the momentum of the electron change if the electron were allowed to be between –0.62 nm and +0.62 nm?

A) The uncertainty in the momentum would be twice its previous value.
B) The uncertainty in the momentum would be half of its previous value.
C) The uncertainty in the momentum would be four times its previous value.
D) The uncertainty in the momentum would be one fourth its previous value.

FROM THE FORMULA SHEET

$$\Delta x \, \Delta p_x \geq \frac{h}{2\pi}$$

Step 1. Δx doubles (0.62 nm to 1.24 nm), so the minimum $\Delta p \propto \frac{1}{\Delta x}$ is halved.

Final answer: **(B) Half its previous value** ✓ Matches answer key

Question 28 75% energy loss in Compton scattering

28. What percentage increase in wavelength leads to a 75% loss of photon energy in a photon-free electron collision?

A) 200 %

B) 300 %

C) 400 %

D) 500 %

FROM THE FORMULA SHEET

$$E = hf$$

REMEMBER — not on the formula sheet

$E = \frac{hc}{\lambda}$, so $\lambda \propto \frac{1}{E}$.

Step 1. After losing 75%: $E' = \frac{E}{4}$, so $\lambda' = 4\lambda$.

$$\frac{\lambda' - \lambda}{\lambda} = 3 = 300\%$$

Final answer: **(B) 300%** ✓ Matches answer key

Question 29 Doubling kinetic energy

29. If you double the kinetic energy of a nonrelativistic particle, how does its de Broglie wavelength change?

A) $1/2$
B) 2
C) $\sqrt{2}$
D) $1/\sqrt{2}$

FROM THE FORMULA SHEET

$p = \dfrac{h}{\lambda}, \quad K = \dfrac{1}{2}mv^2$

REMEMBER — not on the formula sheet

$p = \sqrt{2mK}$, so $\lambda = \dfrac{h}{\sqrt{2mK}} \propto \dfrac{1}{\sqrt{K}}$.

$$\frac{\lambda'}{\lambda} = \sqrt{\frac{K}{2K}} = \frac{1}{\sqrt{2}}$$

Final answer: **(D)** $\frac{1}{\sqrt{2}}$ ✓ Matches answer key

Question 30 Probability in a narrow strip

30. The wave functions for an electron in an infinite, one dimensional potential well with length L along an x axis is given by $\psi(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi}{L}x\right)$. What is the probability of finding the electron between $x = (0.50)L$ and $x + dx = (0.51)L$?

A) 0.005
B) 0.014
C) 0.020
D) 0.050

FROM THE FORMULA SHEET

$p(x) = \psi_n^2(x)\,dx$

Step 1. The strip is narrow ($dx = 0.01L$), so evaluate ψ^2 at $x = 0.50L$:

$$p = \frac{2}{L}\sin^2\left(\frac{\pi}{2}\right)(0.01L) = 2(1)(0.01) = 0.020$$

Final answer: **(C) 0.020** ✓ Matches answer key